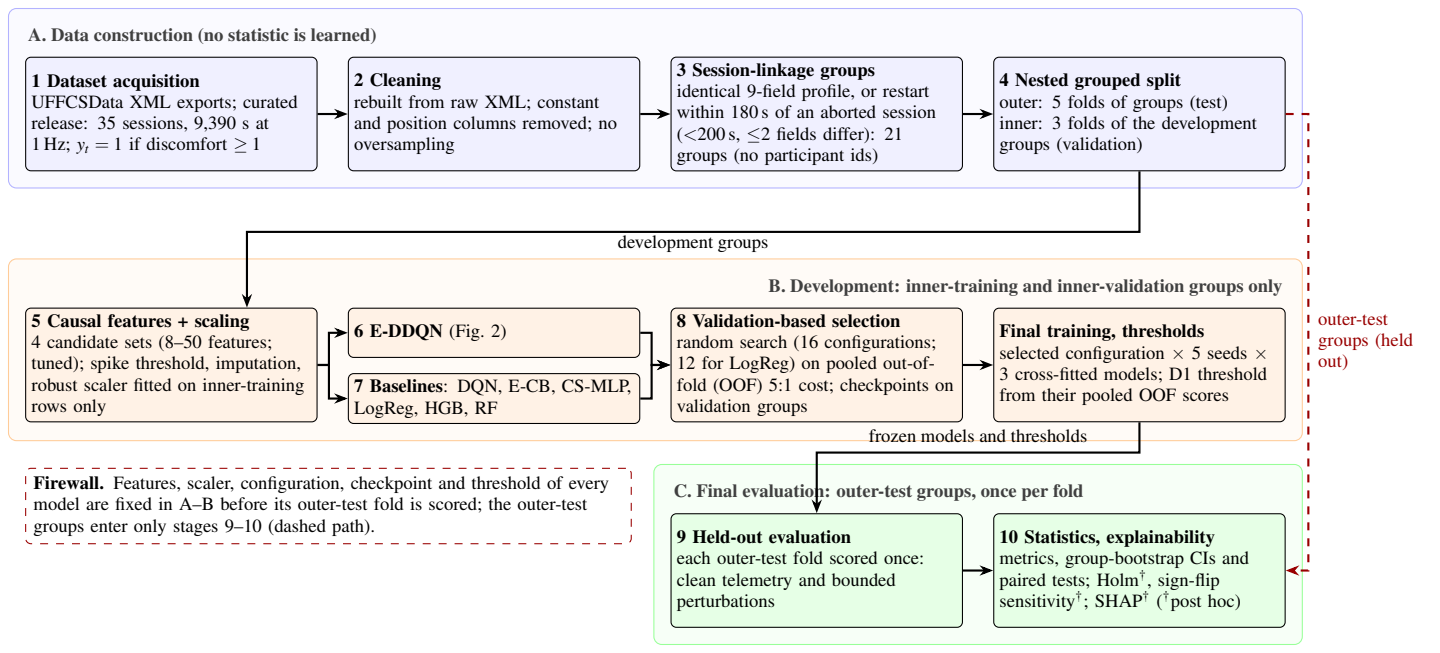

A. Data construction (no statistic is learned)
1 Dataset acquisition
UFFCSData XML exports; curated release: 35 sessions, 9,390 s at 1 Hz; $y_t = 1$ if discomfort ≥ 1
2 Cleaning
rebuilt from raw XML; constant and position columns removed; no oversampling
3 Session-linkage groups
identical 9-field profile, or restart within 180 s of an aborted session (<200 s, ≤2 fields differ): 21 groups (no participant ids)
4 Nested grouped split
outer: 5 folds of groups (test)
inner: 3 folds of the development groups (validation)
development groups
B. Development: inner-training and inner-validation groups only
5 Causal features + scaling
4 candidate sets (8–50 features; tuned); spike threshold, imputation, robust scaler fitted on inner-training rows only
6 E-DDQN (Fig. 2)
7 Baselines: DQN, E-CB, CS-MLP, LogReg, HGB, RF
8 Validation-based selection
random search (16 configurations; 12 for LogReg) on pooled out-of-fold (OOF) 5:1 cost; checkpoints on validation groups
Final training, thresholds
selected configuration × 5 seeds × 3 cross-fitted models; D1 threshold from their pooled OOF scores
outer-test groups (held out)
frozen models and thresholds
Firewall. Features, scaler, configuration, checkpoint and threshold of every model are fixed in A–B before its outer-test fold is scored; the outer-test groups enter only stages 9–10 (dashed path).
C. Final evaluation: outer-test groups, once per fold
9 Held-out evaluation
each outer-test fold scored once: clean telemetry and bounded perturbations
10 Statistics, explainability
metrics, group-bootstrap CIs and paired tests; Holm†; sign-flip sensitivity†; SHAP† (†post hoc)